# Mobile Augmented Geo Data Visualization Based on OSM

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# Objective

- Augmented Reality („When the real world is enriched with virtual content…“)
  - Navigation-aid for hiking and trekking
  - Rendering POIs and paths into the camera‘s live-view in 3D
  - Open Street Map data
  - Proof of concept
  - Android

# AR-Browser

- Wikitude / PeakAR
- Layar
- INSTAR
- …









# Technology

- Basis: PeakAR Salzburg Research
- Master thesis project

salzburg research

- Android (G1 /HTC Dream)
- OpenGL ES
- Sensor data
  - GPS
  - Compass
  - G-force sensor
  - Camera
- Open Street Map / web services





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# Concept

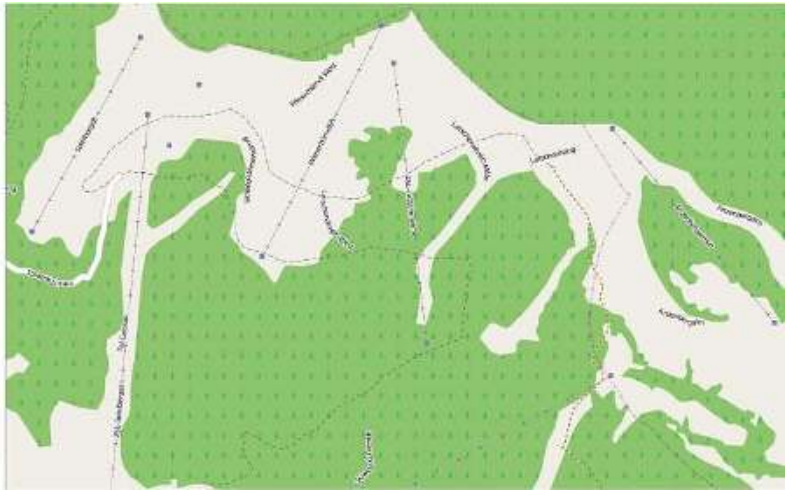
- View vector
  - Position and height from GPS
  - Cardinal direction from compass
  - Tilt from g-force sensor
- OpenGL Models
  - Created from OSM data
  - User is “in the model”
- Correct Perspective!

The data directly influences the accuracy!

# Open Street Map

- Open Street Map
  - High degree of detail
  - Metadata
  - „Free“
  - Access to geo-data
  - No (sufficient) height information?



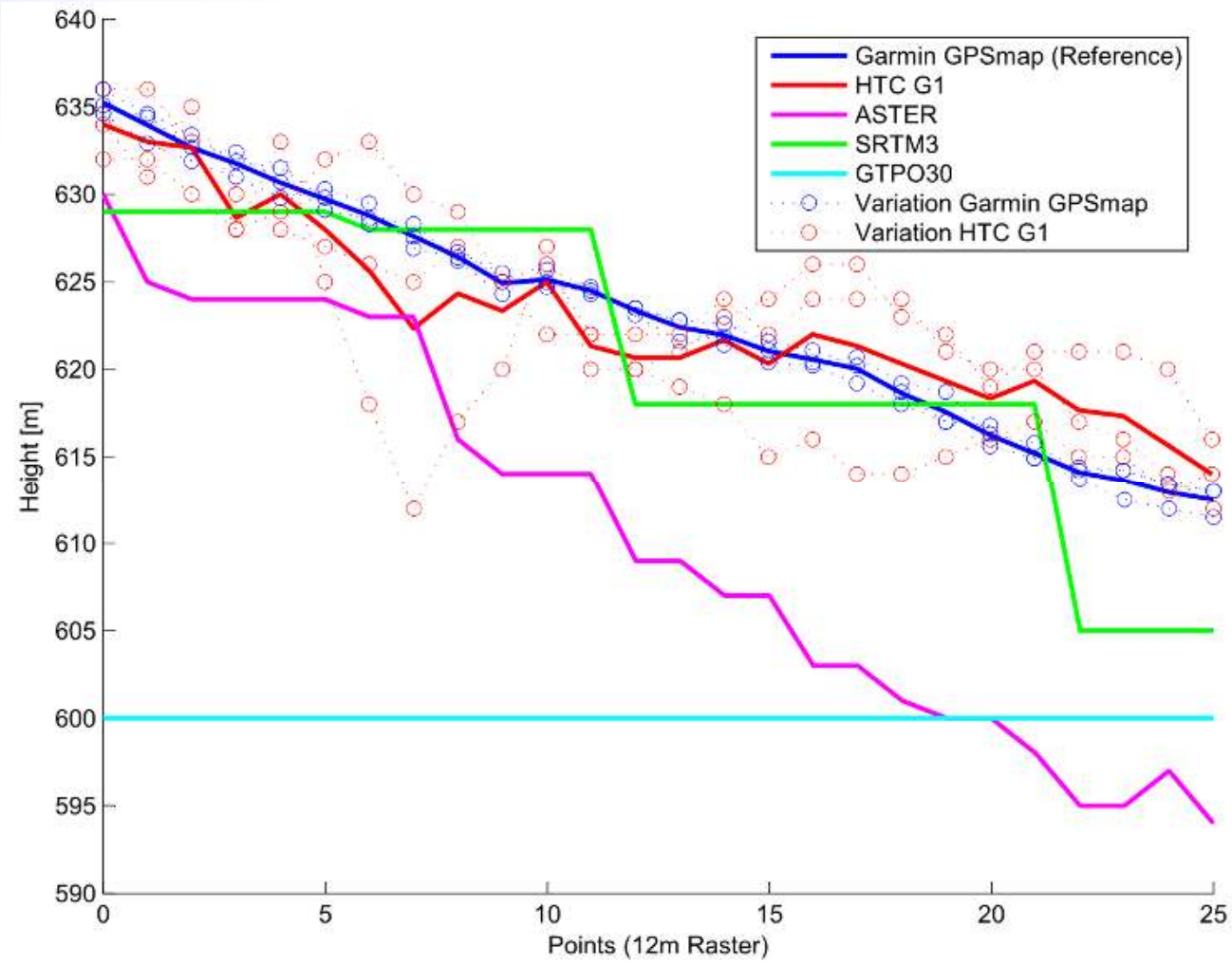



# Height?

- Height is important for the model!
- „Flat“ visualization is not satisfactory
  - Paths winding up a mountain
  - Mountain peaks / Plateau
  - Webservices

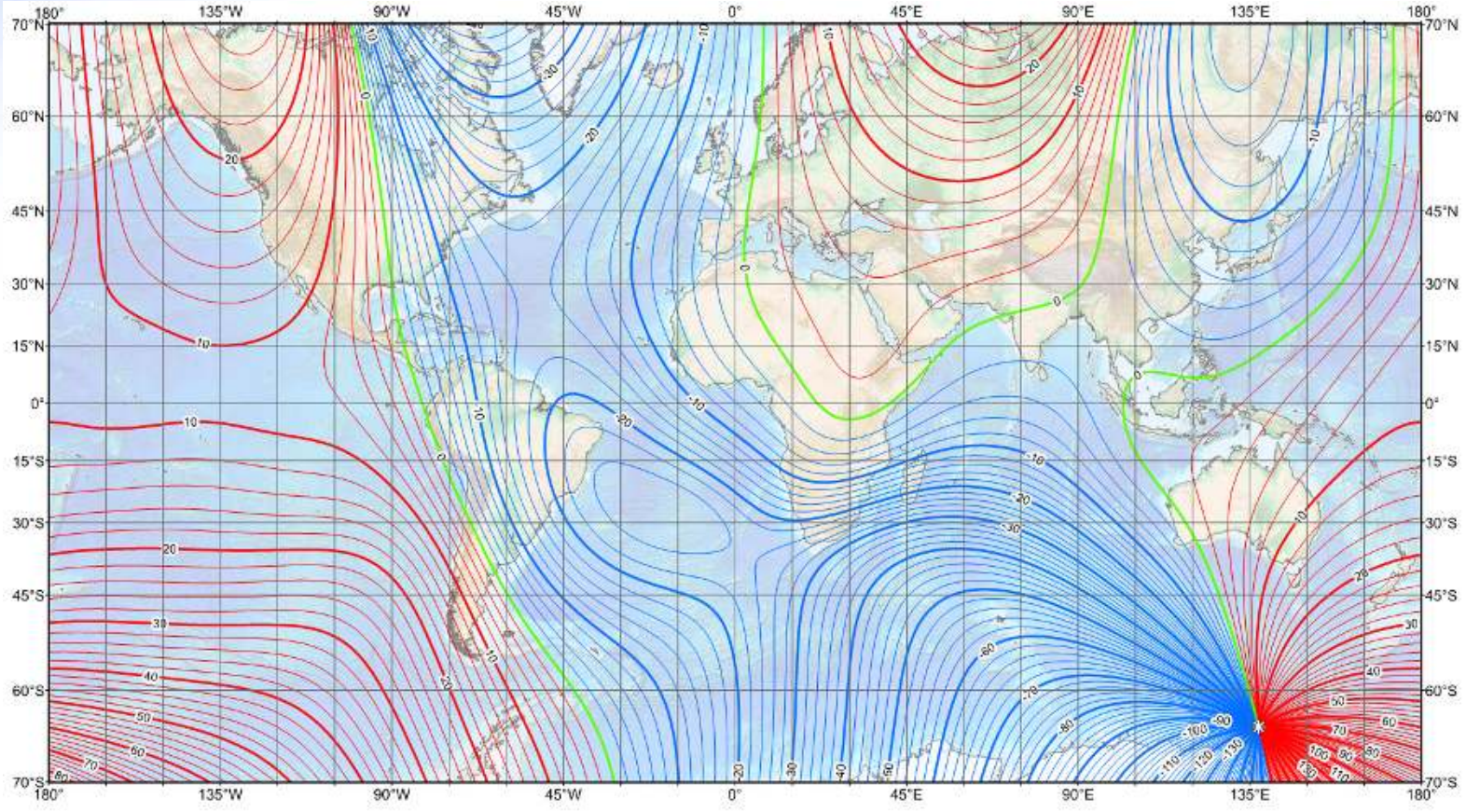
- Height of the device? Use GPS height?
  - Reference system (GPS height vs. national systems)
  - Absolute accuracy not of interest -> consistency!
  - Acquire device height from webservice or use GPS height?

# Topography


Garmin GPSmap (Reference)
HTC G1
ASTER
SRTM3
GTPO30
Variation Garmin GPSmap
Variation HTC G1
Height [m]
640
635
630
625
620
615
610
605
600
595
590
0
5
10
15
20
25
Points (12m Raster)

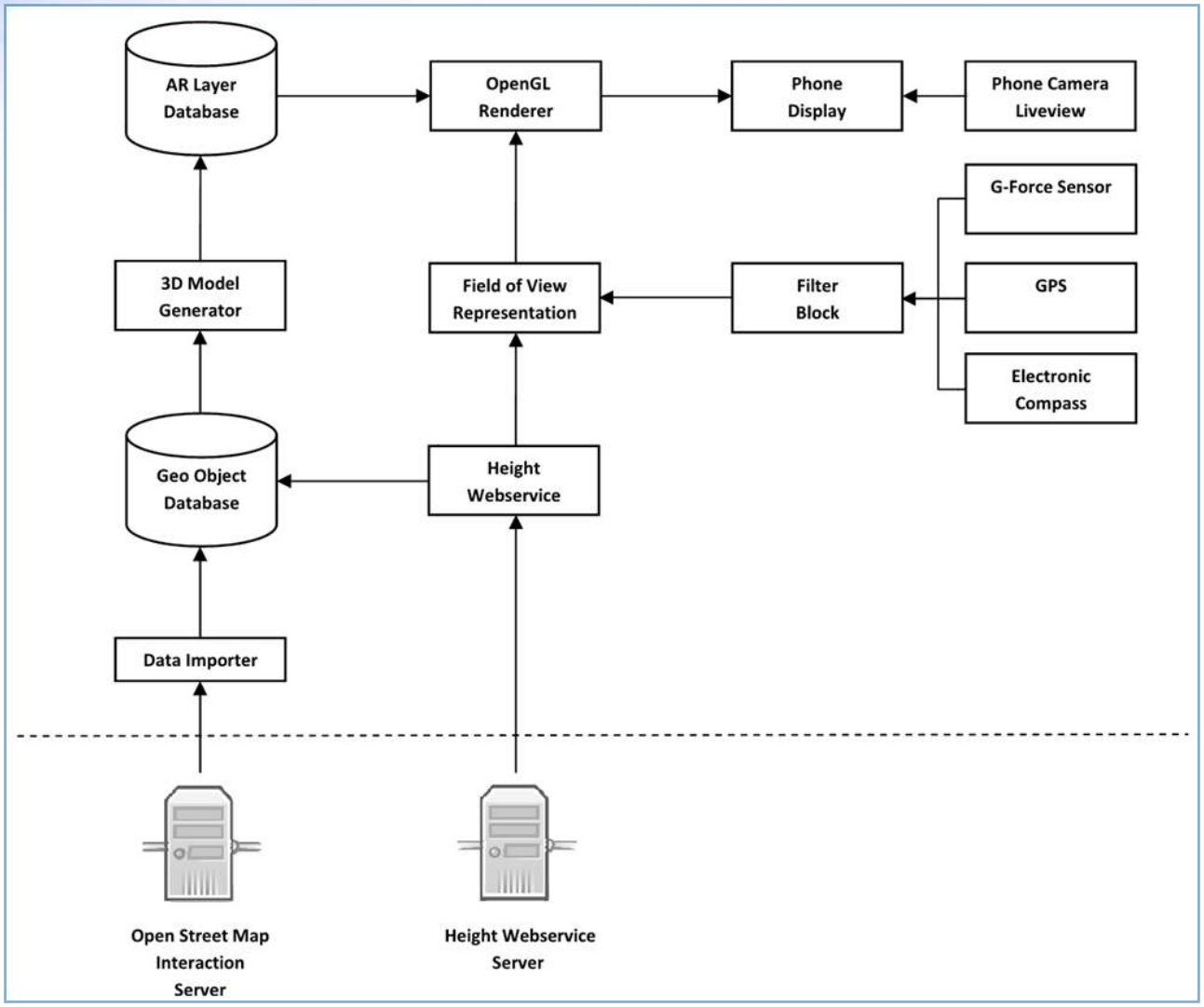
# Heading

- Compass points to magnetic „north“ (not grid north)




World Magnetic Model Epoch 2010 - Main Field Declination (D). Mercator Map Projection with declination in degrees.
Source: http://www.ngdc.noaa.gov

# Concept

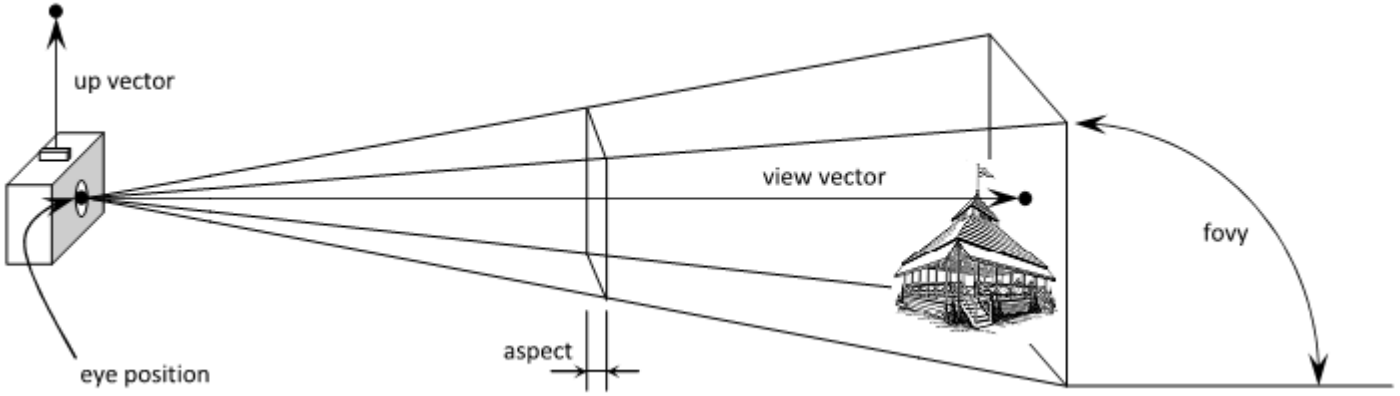

AR Layer Database
OpenGL Renderer
Phone Display
Phone Camera Liveview
G-Force Sensor
3D Model Generator
Field of View Representation
Filter Block
GPS
Electronic Compass
Geo Object Database
Height Webservice
Data Importer
Open Street Map Interaction Server
Height Webservice Server

# Models

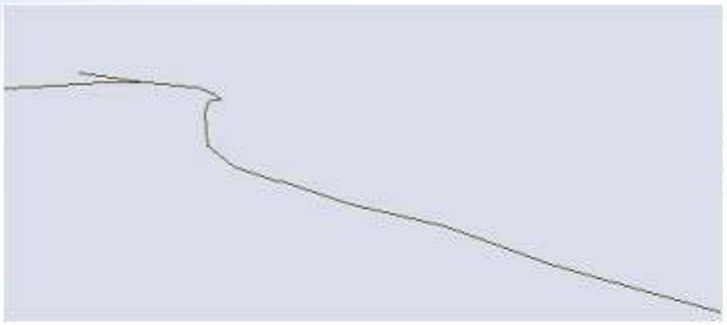
- 3D Model Creator
  - Paths
  - Areas
  - POIs
  - Labels
- Paths and areas consist of points
  -> conversion to vertex models (UTM)
- Text and icons

# Perspective

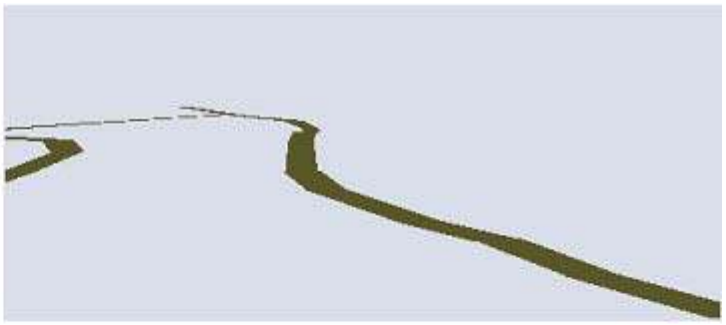
- Perspective Correction
  - Perspective is defined by the camera's focal length
  - Near objects too big – distant disappear (~700m)
    - Text Labels – fixed scaling (counter perspective)
    - POIs – bound scaling (max. and min. size)
    - Paths (2D and 3D mixed model)



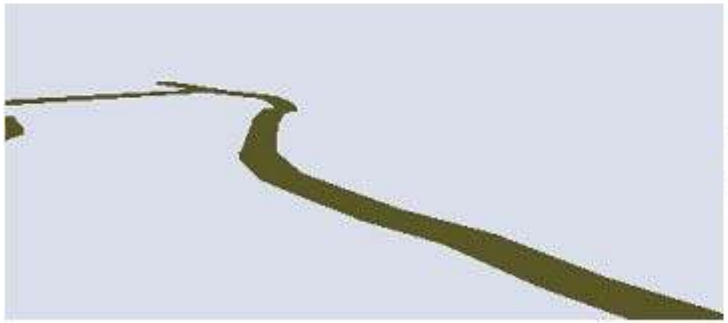
# 3D Model Creator



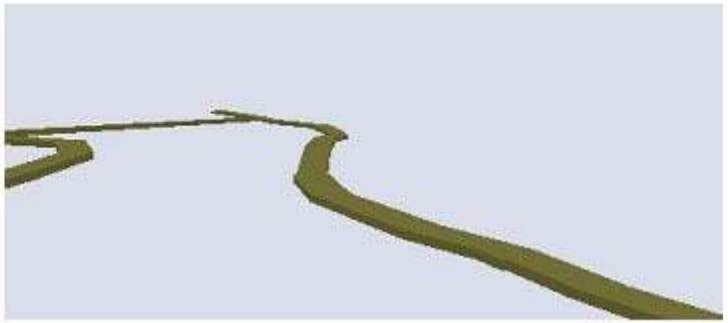
(a) A path line without perspective.



(b) A path having a width but no height.



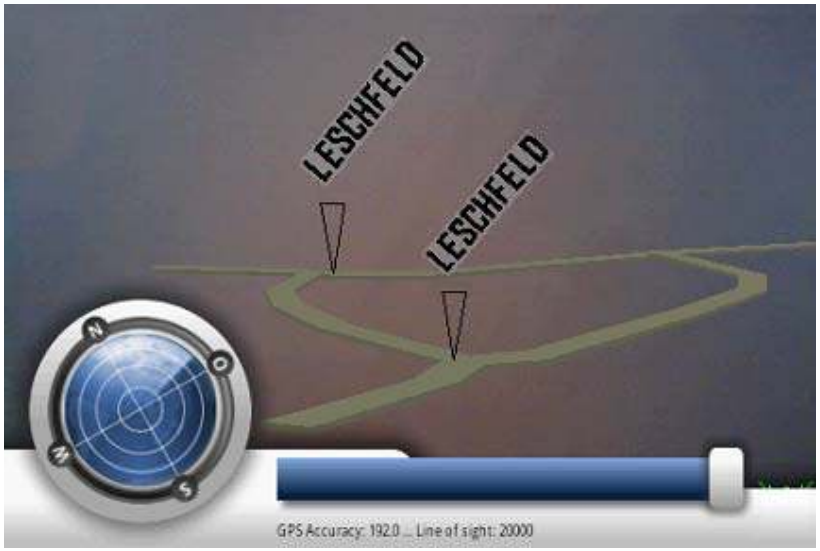
(c) A 3D Path drawn in one color.



(d) A 3D Path with the sides being drawn in a darker color.

# Labels

- Creates 3D labels
- Rotation
- Readability
- Placement (continuous?)

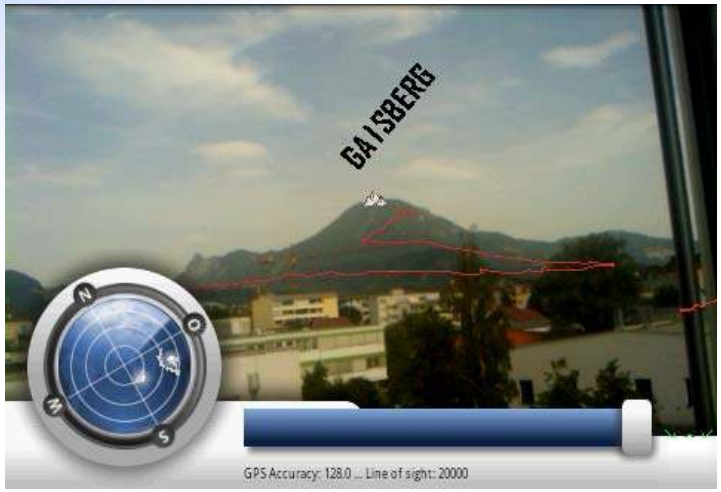






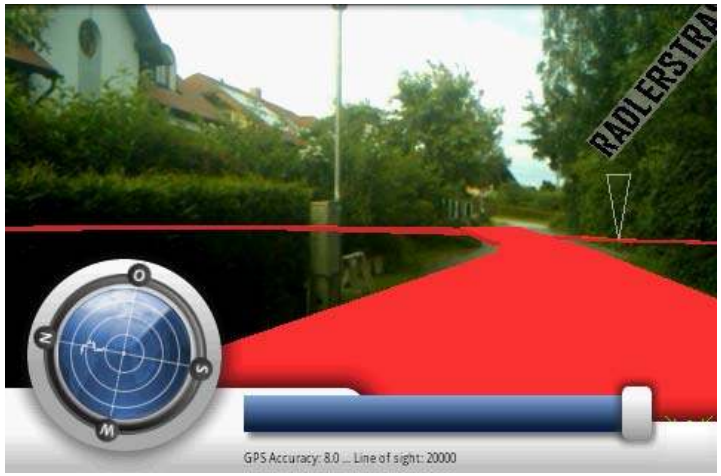


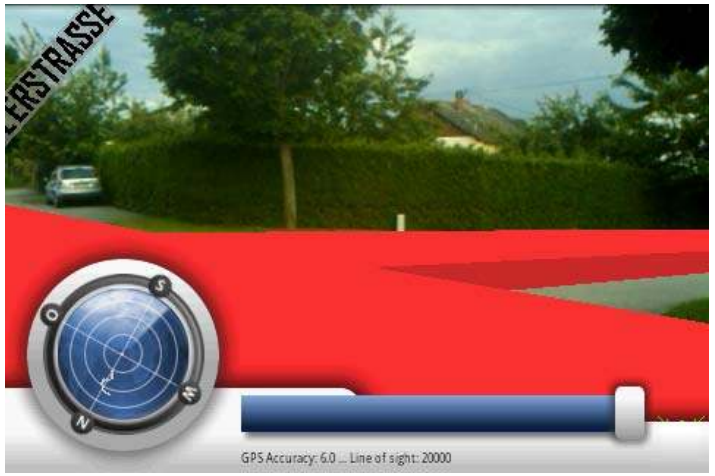
# Screenshots









# Screenshots






# Conclusion

- Scenarios
  - Far (compass, g-force sensor)
    - POI ✓
    - Paths ✓
  - Near (GPS, geo-data, height)
    - POI ✓
    - Paths
      - Accurate data ✓
      - Inaccurate data ⊘
        (User moves along a path -> matching) ○

# Remaining Issues

- Topographic data
  - 3D Mesh for topography
- Dependency towards network connection
  - Compass
  - Topography
  - OSM-Data
- Phone sensor inaccuracies
- Solution? Offline database with pre-calculated models

# Further information

- www.fh-hagenberg.at
- www.salzburgresearch.at
- SotM Paper:
  Mobile Augmented Geo Data Visualization Based on OSM